

FinRAG

An Agentic GraphRAG System for Financial Question Answering

Team C09

Problem Statement

- Financial disclosures (10-K filings, earnings calls) are long, dense, and numerically precise – exactly the setting where plain LLMs and vector-only RAG struggle.
- Multi-hop financial questions require combining facts scattered across **sections, companies, and years** – e.g. “How did Company A’s R&D spend growth compare to Company B’s over the last two fiscal years?”
- Knowledge graphs (KGs) offer a structured, auditable substrate for this kind of reasoning – but **how an autonomous agent should use a financial KG** is still an open, actively evolving question.
- Our project builds on the **FinReflectKG** knowledge graph (17.5M triplets, S&P 500 10-K filings, 2014–2024) and asks: *what is the right agentic architecture to reason over it?*

Problem Statement – Research Gaps from Existing Work

Reviewing the existing literature (detailed next section) surfaces four concrete, evidenced gaps:

1. **Retrieval strategy vs. reasoning-type interaction is underexplored.** The FinReflectKG-MultiHop benchmark shows KG-linked evidence beats page-window retrieval by $\sim 24\%$ accuracy at $\sim 84.5\%$ fewer tokens – but this gain is reported in aggregate, not broken down by *temporal*, *cross-company*, and *intra-document* reasoning types.
2. **No latency-aware adaptive retrieval.** Existing agentic-RAG and GraphRAG systems (HybridRAG, Think-on-Graph) choose a retrieval strategy statically; none adaptively routes “easy” single-hop questions to cheap vector search and reserves expensive graph traversal for genuinely multi-hop questions.
3. **Provenance is present in the data but not evaluated as a system property.** FinReflectKG links triples to exact document chunks, yet no existing agentic financial-QA system scores or audits its own *answer-to-evidence-path* traceability as an explicit metric.
4. **No sensitivity analysis by filing year / training-cutoff proximity.** Since 10-K data spans 2014–2024, filings close to an LLM’s knowledge cutoff risk being contaminated by parametric memory rather than genuine retrieval – a confound no existing benchmark controls for.

Literature Review

1. The benchmark our project builds on: **FinReflectKG-MultiHop**
2. The knowledge graph itself: **FinReflectKG construction**
3. Foundational **GraphRAG** (domain-general)
4. **Agentic RAG** – autonomous tool-use over retrieval
5. **Graph-traversal agents**: Think-on-Graph
6. **Financial-domain GraphRAG**: HybridRAG, FinDKG

1. FinReflectKG-MultiHop (Benchmark)

Arun, Harsh, Sarmah, Pasquali (2025)

<https://arxiv.org/abs/2510.02906>

- Benchmark for multi-hop financial QA over a temporally-indexed KG linking audited triples to source chunks from S&P 100 10-K filings (2022–2024).
- Mines frequent 2–3 hop subgraph patterns (via GICS sector taxonomy) to generate analyst-style questions with exact KG-grounded evidence.
- Three retrieval scenarios compared: **S1** precise KG-linked paths, **S2** text-only page windows, **S3** page windows with distractors/randomization.
- **Key finding:** KG-guided evidence improves correctness by $\sim 24\%$ while cutting token usage by $\sim 84.5\%$ vs. page-window retrieval, consistent across reasoning and non-reasoning LLMs and across two different judge models.
- Question types span **intra-document**, **inter-year**, and **cross-company** relationships – this is exactly the axis our Gap #1 asks to be evaluated in isolation, not only in aggregate.

2. FinReflectKG (Knowledge Graph Construction)

Arun, Dimino, Agarwal, Sarmah, Pasquali (2025)

<https://arxiv.org/abs/2508.17906>

- Introduces the open-source financial KG our project uses (S&P 100/500 10-K filings; our subset: 17.5M triplets, 2014–2024).
- Construction pipeline: intelligent document parsing → table-aware chunking → schema-guided iterative extraction → **reflection-driven feedback loop**.
- Three extraction modes compared – single-pass, multi-pass, reflection-agent-based; the reflection-agent mode wins, reaching **64.8% rule compliance** (a 53.2% relative gain over single-pass).
- Evaluation pipeline combines rule-based checks, statistical validation, and LLM-as-a-Judge scoring.
- **Relevance to us:** the KG is a given resource, not something we build – but understanding its reflection-driven provenance is essential context for how trustworthy our agent's retrieved evidence actually is.

3. GraphRAG – Foundational (Domain-General)

Edge, Trinh, Cheng, et al. (Microsoft, 2024)

<https://arxiv.org/abs/2404.16130>

- Introduces **GraphRAG**: build an LLM-derived entity knowledge graph from source text, partition it into hierarchical communities (Leiden algorithm), then pre-generate community summaries.
- At query time, each relevant community summary produces a partial answer; partial answers are map-reduced into a final response.
- Targets *global sensemaking* questions (“what are the main themes...”) that vector RAG cannot answer, since these require synthesis across the whole corpus, not a single passage.
- Shows substantial gains over naive vector RAG in comprehensiveness and answer diversity on million-token-scale corpora.
- **Relevance to us**: this is the architectural ancestor of graph-based retrieval generally – but it is community-summary based and domain-general, not built for structured, audited, numerically precise financial triples.

4. Agentic RAG – Autonomous Tool-Use

Singh et al. (2025), Survey

<https://arxiv.org/abs/2501.09136>

- Surveys the shift from static RAG pipelines to **Agentic RAG**: LLM-driven agents that plan, reflect, select tools, and adapt retrieval strategy dynamically rather than following one fixed pipeline.
- Introduces a taxonomy along agent cardinality (single- vs. multi-agent), control structure, autonomy level, and knowledge representation (including graph-based agentic RAG).
- Surveys applications across healthcare, finance, education, and enterprise document processing.
- **Relevance to us**: directly motivates our project's core design choice – a 9-tool LLM-driven agent, rather than a single fixed retrieval pipeline, deciding *which* tool (graph traversal, hybrid search, symbolic computation, etc.) to invoke per sub-question.

5. Think-on-Graph – Agentic Graph Traversal

Sun et al. (2023)

<https://arxiv.org/abs/2307.07697>

- Proposes the “LLM $\otimes$ KG” paradigm: the LLM acts as an **agent that interactively explores** entities and relations directly on the KG, rather than retrieving a static subgraph once.
- Implements this via **Think-on-Graph (ToG)**: iterative beam search over the KG – at each step, the LLM decides which relations/entities to explore next (*explore*) and which reasoning paths to keep (*prune*).
- Produces explicit, editable multi-hop reasoning paths – improving both depth of reasoning and explainability/traceability of the final answer.
- **Relevance to us:** this is the closest prior work to our “graph traversal” tool – but ToG is evaluated on general-domain KGs (e.g. Wikidata-style facts), not on audited financial disclosures with numerically precise, time-indexed triples.

6. HybridRAG – Financial-Domain GraphRAG

Sarmah, Hall, Rao, Patel, Pasquali, Mehta (2024)

<https://arxiv.org/abs/2408.04948>

- Directly targets financial text: earnings call transcripts (Nifty 50 companies), which are domain-specific, numerically dense, and Q&A-structured.
- Combines **VectorRAG** + **GraphRAG** into a single **HybridRAG** pipeline, retrieving context from both a vector database and a constructed KG simultaneously.
- Shown to outperform either VectorRAG or GraphRAG alone, on both retrieval accuracy and answer-generation quality, using ground-truth Q&A pairs naturally present in earnings call transcripts.
- **Relevance to us:** strongest existing evidence that hybrid (graph + vector) retrieval specifically helps in *financial* text – our ablation ladder (no-retrieval → vector RAG → graph-enriched RAG → agentic GraphRAG) is designed to isolate exactly how much of that gain an autonomous agent can additionally capture over a fixed hybrid pipeline.

6b. FinDKG – Dynamic Financial Knowledge Graphs

Li & Passino (2024)

<https://arxiv.org/abs/2407.10909>

- Builds an open-source **dynamic** KG (FinDKG) from a corpus of financial news articles using an LLM-based extraction pipeline (ICKG).
- Proposes **KGTransformer**, an attention-based GNN for link prediction and trend detection over the evolving graph.
- Applied to thematic investing – shown to outperform existing thematic ETFs on the tested period.
- **Relevance to us:** demonstrates the value of *temporal* structure in financial KGs for forecasting-style tasks; reinforces why our benchmark's inter-year / temporal question type matters, though FinDKG targets prediction rather than question answering.

Literature Review – Summary Table

Paper	Core Contribution	Gap vs. Our Work
FinReflectKG-MultiHop	KG-linked evidence beats page-window retrieval (+24% acc, −84.5% tokens)	No per-reasoning-type breakdown; no agentic tool selection
FinReflectKG	Reflection-driven agentic KG construction pipeline	KG is given to us, not built; used as provenance context
GraphRAG (MSFT)	Community-summary graph index for global sensemaking	Domain-general, not numerically precise / audited
Agentic RAG Survey	Taxonomy of planning / tool-use / reflection agents	Motivates but doesn't specify a financial 9-tool agent
Think-on-Graph	Iterative beam-search KG traversal agent	General-domain KGs, no financial numeric grounding
HybridRAG	Vector+Graph hybrid beats either alone	Fixed pipeline, not an

Our Novelty

Where We Sit Relative to Prior Work

- The KG (FinReflectKG) and the benchmark (FinReflectKG-MultiHop) are **given** – our contribution is the **agent system** built on top of them, evaluated via a controlled ablation ladder:

no-retrieval → vector RAG → graph-enriched RAG → **agentic GraphRAG**

- Core system: a **9-tool LLM-driven agent** (graph traversal, hybrid passage search, symbolic/numerical computation, and further tools), choosing which tool to invoke per sub-question rather than following one fixed retrieval path.
- **Important scoping note:** a near-complete, independently authored paper (Capozzi & Helbing, ETH Zürich, expected 2026) has since emerged with substantial architectural overlap – same KG, same benchmark, a similar 9-tool structure and ablation design. Our genuine remaining novelty is therefore concentrated in evaluation axes that paper does not appear to cover, listed next.

Novelty #1 & #2: Fine-Grained Accuracy & Adaptive Routing

1. Fine-grained reasoning-type accuracy splits

Rather than reporting one aggregate accuracy number, evaluate the agent **separately** on temporal (inter-year), cross-company, and intra-document questions. Directly answers: *does agentic tool selection help uniformly, or mainly on one reasoning type?*

2. Latency-aware adaptive tool routing

Add a routing layer that estimates question complexity up front and skips expensive graph traversal for simple, single-hop lookups – trading a small accuracy cost (if any) for a measurable latency/token reduction on the easy end of the distribution. No existing agentic financial-QA system in our review does this adaptively.

Novelty #3 & #4: Provenance Auditing & Cutoff Sensitivity

3. Provenance / audit-trail evaluation

FinReflectKG already links every triple to an exact filing, page, and chunk. We propose scoring the agent's **final answer against its own retrieved evidence path** as an explicit metric (can every claim in the answer be traced to a specific retrieved triple/chunk?) – treating traceability as a first-class evaluation axis, not just a nice property of the underlying data.

4. Training-cutoff sensitivity by filing year

Since our data spans 2014–2024 and LLMs have their own knowledge cutoffs, we propose deliberately splitting evaluation by filing year relative to each tested model's cutoff – to check whether reported gains hold for *recent* filings the model could not have memorized, or shrink once the KG is the only source of truth.

Dataset

Dataset – FinReflectKG

- **Source:** domyn/FinReflectKG on Hugging Face, CC-BY-NC-4.0 license.
- **Scale:** 17.5M triplets extracted from S&P 500 SEC 10-K filings, spanning **2014–2024**.
- **Structure:** each triple is linked to an exact evidence chunk (filing, page, chunk ID) – giving built-in provenance, which our Novelty #3 evaluation directly exploits.
- **Our working subset:** initial data-loading pass filters to **AAPL and MSFT** (via the lowercase ticker column), loaded together into a **Neo4j AuraDB** (free tier) instance, chosen for ease of team collaboration over a local graph database setup.
- **Benchmark pairing:** evaluated against **FinReflectKG-MultiHop**, which provides gold QA pairs with labeled reasoning type (intra-document / inter-year / cross-company) and full KG-linked supporting evidence – exactly the labels our fine-grained accuracy-split novelty needs.

- **Problem:** multi-hop financial QA needs structured, auditable retrieval – KGs help, but the right *agentic* architecture over a financial KG is still underexplored.
- **Literature:** benchmark (FinReflectKG-MultiHop) → KG construction (FinReflectKG) → general GraphRAG → agentic RAG → graph-traversal agents (ToG) → financial GraphRAG (HybridRAG, FinDKG).
- **Novelty:** fine-grained reasoning-type accuracy, latency-aware adaptive routing, provenance/audit-trail scoring, training-cutoff sensitivity by filing year.
- **Dataset:** FinReflectKG (17.5M triplets, S&P 500, 2014–2024), AAPL/MSFT working subset in Neo4j AuraDB.
- *Methodology and system architecture details to follow in the next phase.*

References

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